

CS23532-COMPUTER NETWORKS-LAB MANUAL

Practical -1

AIM: - Study of various Network commands used in Linux and Windows:

Windows commands :

D:\>hostname

LAPTOP-O0K6G7B7

D:\>ipconfig/all

Windows IP Configuration

Host Name : LAPTOP-O0K6G7B7

Primary Dns Suffix :

Node Type : Mixed

IP Routing Enabled. : No

WINS Proxy Enabled. : No

Ethernet adapter Ethernet 4:

Connection-specific DNS Suffix . :

Description : VirtualBox Host-Only Ethernet Adapter

Physical Address. : 0A-00-27-00-00-02

DHCP Enabled. : No

Autoconfiguration Enabled : Yes

Link-local IPv6 Address : fe80::1da4:d7dd:72a8:cf41%2(Preferred)

IPv4 Address. : 192.168.56.1(Preferred)

Subnet Mask : 255.255.255.0

Default Gateway :

DHCPv6 IAID : 352976935

DHCPv6 Client DUID. : 00-01-00-01-2A-C9-4C-BB-90-E8-68-14-92-7B

NetBIOS over Tcpip. : Enabled

Wireless LAN adapter Local Area Connection* 9:

Media State : Media disconnected

Connection-specific DNS Suffix . :

Description : Microsoft Wi-Fi Direct Virtual Adapter

Physical Address. : 92-E8-68-14-92-2B

DHCP Enabled. : Yes

Autoconfiguration Enabled : Yes

Wireless LAN adapter Local Area Connection* 10:

Media State : Media disconnected

Connection-specific DNS Suffix . :

Description : Microsoft Wi-Fi Direct Virtual Adapter #2

Physical Address. : 92-E8-68-14-92-3B

DHCP Enabled. : Yes

Autoconfiguration Enabled : Yes

Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . :

Description : VMware Virtual Ethernet Adapter for VMnet1

Physical Address. : 00-50-56-C0-00-01

DHCP Enabled. : Yes

Autoconfiguration Enabled : Yes

Link-local IPv6 Address : fe80::86fd:79b9:4558:7496%5(Preferred)

IPv4 Address. : 192.168.147.1(Preferred)

Subnet Mask : 255.255.255.0

Lease Obtained. : 31 August 2026 09:52:47

Lease Expires : 31 August 2026 10:22:48

Default Gateway :

DHCP Server : 192.168.147.254

DHCPv6 IAID : 1023430742

DHCPv6 Client DUID. : 00-01-00-01-2A-C9-4C-BB-90-E8-68-14-92-7B

NetBIOS over Tcpip. : Enabled

Ethernet adapter VMware Network Adapter VMnet8:

Connection-specific DNS Suffix . :

Description : VMware Virtual Ethernet Adapter for VMnet8

Physical Address. : 00-50-56-C0-00-08

DHCP Enabled. : Yes

Autoconfiguration Enabled : Yes

Link-local IPv6 Address : fe80::c2e3:901d:6ad1:4bcb%13(Preferred)

IPv4 Address. : 192.168.75.1(Preferred)

Subnet Mask : 255.255.255.0

Lease Obtained. : 31 August 2026 09:52:52

Lease Expires : 31 August 2026 10:22:48

Default Gateway :

DHCP Server : 192.168.75.254

DHCPv6 IAID : 1056985174

DHCPv6 Client DUID. : 00-01-00-01-2A-C9-4C-BB-90-E8-68-14-92-7B

Primary WINS Server : 192.168.75.2

NetBIOS over Tcpip. : Enabled

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :

Description : MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter

Physical Address. : 90-E8-68-14-92-7B

DHCP Enabled. : Yes

Autoconfiguration Enabled : Yes

Link-local IPv6 Address : fe80::ed17:b200:1378:289f%15(Preferred)
IPv4 Address. : 172.30.4.22(Preferred)
Subnet Mask : 255.255.224.0
Lease Obtained. : 31 August 2026 08:45:45
Lease Expires : 01 September 2026 08:45:44
Default Gateway : 172.30.0.1
DHCP Server : 172.30.0.1
DHCPv6 IAID : 227600488
DHCPv6 Client DUID. : 00-01-00-01-2A-C9-4C-BB-90-E8-68-14-92-7B
DNS Servers : 8.8.8.8
NetBIOS over Tcpip. : Enabled
Ethernet adapter Bluetooth Network Connection:
Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Bluetooth Device (Personal Area Network)
Physical Address. : 90-E8-68-14-92-7A
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

D:\>arp -a

Interface: 192.168.56.1 --- 0x2

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.147.1 --- 0x5

Internet Address	Physical Address	Type
192.168.147.254	00-50-56-e9-44-86	dynamic
192.168.147.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.75.1 --- 0xd

Internet Address	Physical Address	Type
192.168.75.254	00-50-56-e6-63-f5	dynamic
192.168.75.255	ff-ff-ff-ff-ff-ff	static

224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 172.30.4.22 --- 0xf

Internet Address	Physical Address	Type
172.30.0.1	cc-82-7f-30-b5-a3	dynamic
172.30.31.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.102.18	01-00-5e-7f-66-12	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

D:\>netstat -r

Interface List

2...0a 00 27 00 00 02VirtualBox Host-Only Ethernet Adapter

4...92 e8 68 14 92 2bMicrosoft Wi-Fi Direct Virtual Adapter

17...92 e8 68 14 92 3bMicrosoft Wi-Fi Direct Virtual Adapter #2

5...00 50 56 c0 00 01VMware Virtual Ethernet Adapter for VMnet1

13...00 50 56 c0 00 08VMware Virtual Ethernet Adapter for VMnet8

15...90 e8 68 14 92 7bMediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter

18...90 e8 68 14 92 7aBluetooth Device (Personal Area Network)

1.....Software Loopback Interface 1

IPv4 Route Table

Active Routes:

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	172.30.0.1	172.30.4.22	35
127.0.0.0	255.0.0.0	On-link	127.0.0.1	331
127.0.0.1	255.255.255.255	On-link	127.0.0.1	331
127.255.255.255	255.255.255.255	On-link	127.0.0.1	331
172.30.0.0	255.255.224.0	On-link	172.30.4.22	291
172.30.4.22	255.255.255.255	On-link	172.30.4.22	291
172.30.31.255	255.255.255.255	On-link	172.30.4.22	291
192.168.56.0	255.255.255.0	On-link	192.168.56.1	281
192.168.56.1	255.255.255.255	On-link	192.168.56.1	281
192.168.56.255	255.255.255.255	On-link	192.168.56.1	281
192.168.75.0	255.255.255.0	On-link	192.168.75.1	291
192.168.75.1	255.255.255.255	On-link	192.168.75.1	291
192.168.75.255	255.255.255.255	On-link	192.168.75.1	291

192.168.147.0	255.255.255.0	On-link	192.168.147.1	291
192.168.147.1	255.255.255.255	On-link	192.168.147.1	291
192.168.147.255	255.255.255.255	On-link	192.168.147.1	291
224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
224.0.0.0	240.0.0.0	On-link	192.168.147.1	291
224.0.0.0	240.0.0.0	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	192.168.75.1	291
224.0.0.0	240.0.0.0	On-link	172.30.4.22	291
255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
255.255.255.255	255.255.255.255	On-link	192.168.147.1	291
255.255.255.255	255.255.255.255	On-link	192.168.56.1	281
255.255.255.255	255.255.255.255	On-link	192.168.75.1	291
255.255.255.255	255.255.255.255	On-link	172.30.4.22	291

Persistent Routes:

None

IPv6 Route Table

Active Routes:

	If Metric	Network Destination	Gateway
1	331	::1/128	On-link
5	291	fe80::/64	On-link
2	281	fe80::/64	On-link

13 291 fe80::/64 On-link

15 291 fe80::/64 On-link

2 281 fe80::1da4:d7dd:72a8:cf41/128

On-link

5 291 fe80::86fd:79b9:4558:7496/128

On-link

13 291 fe80::c2e3:901d:6ad1:4bcb/128

On-link

15 291 fe80::ed17:b200:1378:289f/128

On-link

1 331 ff00::/8 On-link

5 291 ff00::/8 On-link

2 281 ff00::/8 On-link

13 291 ff00::/8 On-link

15 291 ff00::/8 On-link

Persistent Routes:

None

D:\>nslookup www.google.com

Server: dns.google

Address: 8.8.8.8

Non-authoritative answer:

Name: www.google.com

Addresses: 2001:4860:4828:7700::

2001:4860:4827:7700::

2001:4860:482b:7700::

2001:4860:482d:7700::

2001:4860:482a:7700::

2001:4860:482c:7700::

2001:4860:4826:7700::

2001:4860:4829:7700::

142.251.153.119

142.251.156.119

142.251.154.119

142.251.150.119

142.251.157.119

142.251.152.119

142.251.151.119

142.251.155.119

D:\>ping google.com

Pinging google.com [142.250.66.14] with 32 bytes of data:

Reply from 142.250.66.14: bytes=32 time=5ms TTL=118

Reply from 142.250.66.14: bytes=32 time=6ms TTL=118

Reply from 142.250.66.14: bytes=32 time=23ms TTL=118

Reply from 142.250.66.14: bytes=32 time=26ms TTL=118

Ping statistics for 142.250.66.14:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 5ms, Maximum = 26ms, Average = 15ms

D:\>route print

Interface List

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1.....Software Loopback Interface 1

IPv4 Route Table

Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
	0.0.0.0	0.0.0.0	172.30.0.1	172.30.4.22	35
	127.0.0.0	255.0.0.0	On-link	127.0.0.1	331
	127.0.0.1	255.255.255.255	On-link	127.0.0.1	331
127.255.255.255	255.255.255.255		On-link	127.0.0.1	331
	172.30.0.0	255.255.224.0	On-link	172.30.4.22	291
	172.30.4.22	255.255.255.255	On-link	172.30.4.22	291
	172.30.31.255	255.255.255.255	On-link	172.30.4.22	291
	192.168.56.0	255.255.255.0	On-link	192.168.56.1	281
	192.168.56.1	255.255.255.255	On-link	192.168.56.1	281
	192.168.56.255	255.255.255.255	On-link	192.168.56.1	281
	192.168.75.0	255.255.255.0	On-link	192.168.75.1	291
	192.168.75.1	255.255.255.255	On-link	192.168.75.1	291
	192.168.75.255	255.255.255.255	On-link	192.168.75.1	291
	192.168.147.0	255.255.255.0	On-link	192.168.147.1	291
	192.168.147.1	255.255.255.255	On-link	192.168.147.1	291
	192.168.147.255	255.255.255.255	On-link	192.168.147.1	291
	224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
	224.0.0.0	240.0.0.0	On-link	192.168.147.1	291
	224.0.0.0	240.0.0.0	On-link	192.168.56.1	281
	224.0.0.0	240.0.0.0	On-link	192.168.75.1	291

224.0.0.0	240.0.0.0	On-link	172.30.4.22	291
255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
255.255.255.255	255.255.255.255	On-link	192.168.147.1	291
255.255.255.255	255.255.255.255	On-link	192.168.56.1	281
255.255.255.255	255.255.255.255	On-link	192.168.75.1	291
255.255.255.255	255.255.255.255	On-link	172.30.4.22	291

Persistent Routes:

None

IPv6 Route Table

Active Routes:

If	Metric	Network	Destination	Gateway
1	331	::1/128		On-link
5	291	fe80::/64		On-link
2	281	fe80::/64		On-link
13	291	fe80::/64		On-link
15	291	fe80::/64		On-link
2	281	fe80::1da4:d7dd:72a8:cf41/128		
				On-link
5	291	fe80::86fd:79b9:4558:7496/128		
				On-link
13	291	fe80::c2e3:901d:6ad1:4bcb/128		

On-link

15 291 fe80::ed17:b200:1378:289f/128

On-link

1 331 ff00::/8 On-link

5 291 ff00::/8 On-link

2 281 ff00::/8 On-link

13 291 ff00::/8 On-link

15 291 ff00::/8 On-link

Persistent Routes:

None

D:\>tracert google.com

Tracing route to google.com [142.250.66.14]

over a maximum of 30 hops:

1 26 ms 1 ms 1 ms 172.30.0.1

2 19 ms 2 ms 3 ms 137.97.163.129

3 * * * Request timed out.

4 10 ms 8 ms 5 ms 172.16.12.64

5 9 ms 6 ms 6 ms 172.16.12.64

6 44 ms 6 ms 5 ms 72.14.215.136

7 6 ms 9 ms 6 ms 142.251.70.199

8 48 ms 6 ms 6 ms 142.251.60.187

9 24 ms 6 ms 6 ms bom07s35-in-f14.1e100.net [142.250.66.14]

Trace complete.

Linux commands :

traceroute google.com

\$ traceroute google.com

traceroute to google.com (142.250.183.14), 30 hops max, 60 byte packets

1 192.168.1.1 2.123 ms 1.987 ms 2.015 ms

2 10.20.0.1 8.432 ms 8.210 ms 8.345 ms

3 100.64.0.1 15.234 ms 15.102 ms 15.187 ms

4 142.250.183.14 25.456 ms 25.321 ms 25.402 ms

ip addr

\$ ip addr

1: lo: <LOOPBACK,UP,LOWER_UP>

inet 127.0.0.1/8 scope host lo

2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP>

inet 192.168.1.10/24 brd 192.168.1.255

inet6 fe80::1234:5678:abcd:ef01/64

netstat -ano

\$ netstat -ano

Active Internet connections

Proto	Local Address	Foreign Address	State
tcp	0.0.0.0:22	0.0.0.0:*	LISTEN
tcp	192.168.1.10:45678	142.250.183.14:443	ESTABLISHED
tcp	192.168.1.10:52341	142.250.72.14:443	ESTABLISHED

ip route

\$ ip route

default via 192.168.1.1 dev eth0

192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.10